

Evolutionary Computing (X_400111)

Standard Assignment, Task II: Neuroevolution

Send the Best, the Worst, or Anyone?

Emigrant Selection in Island-Model Neuroevolution
of a Walking Robot

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ABSTRACT

We evolve the 212 weights of a 16-8-4-8 neural network that makes the eight-hinge spider_8 robot of the ARIEL framework walk 2 m to a target on the OlympicArena within 15 s. Our research question is how the emigrant-selection policy of a four-island ring model (migrating the best, the worst or random individuals, or not migrating at all) affects the speed and the outcome of evolution. Over five paired seeds of 12,000 evaluations each, every evolutionary algorithm beats random search (final fitness 0.96–1.25 against 2.60). Migration of any kind speeds up convergence compared with isolated islands (posterior probability 89–96% on the area under the convergence curve), and migrating the best is the fastest: it reached a fitness of 1.6 after 2,906 evaluations on average, against 5,466 without migration. The final fitness and the performance on 20 unseen arenas, however, do not depend measurably on the policy, and a single panmictic population was as fast and ended best. A follow-up shows that *how often* the islands migrate matters more than *whom* they send: every 5–20 generations is equally fast, every 50 is as slow as no migration. Finally, the 15 s limit binds: given 20 s, 11 of 30 evolved brains reach their target instead of 3.

KEYWORDS

Neuroevolution, island model, migration policy, evolutionary robotics, legged locomotion

1 INTRODUCTION

Evolutionary robotics designs robot controllers by evolving them in simulation [11]. A common choice is to fix the topology of a neural network and evolve its weights [7]: a continuous search space of hundreds of dimensions, rugged and multimodal, where a population easily converges early on a mediocre gait.

Island models are a standard defence against such premature convergence. The population is split into sub-populations (islands, or demes) that evolve separately and exchange a few individuals from time to time (migration) [1, 3, 16]. Isolation lets the islands explore different regions; migration spreads good genetic material. How an island model behaves depends on the topology, the interval and size of migration, how emigrants are selected and whom they replace [13]. Cantú-Paz [4] showed that choosing emigrants by fitness raises the selection pressure of the whole algorithm, which speeds up convergence but risks taking every island to the same solution. Those results concern genetic algorithms on benchmark functions. Neuroevolution adds a twist: two networks that solve a task can order their hidden neurons differently (the competing-conventions problem [12]), so crossing a native with an immigrant from another lineage can produce a broken child.

Whether migrating the best is still the right choice for evolving a walking robot is therefore an open, practical question.

We study it on the task of the assignment: evolve the weights of a neural-network controller so that a fixed robot body walks from its spawn point to a target. Our research question is:

RQ. How does the emigrant-selection policy of an island-model EA (migrating the *best*, the *worst* or *random* individuals, or *no* migration) affect the convergence speed and the final quality of evolved locomotion controllers?

Following Cantú-Paz [4], we expect: **H1**, migration of any kind converges faster than isolated islands; **H2**, migrating the best converges fastest, but ends at an equal or worse plateau than random migration, because the islands lose their differences; **H3**, migrating the worst behaves close to no migration. We compare the four policies with a standard single-population EA and random search, on five paired seeds each, and follow up with the migration interval and with longer test walks.

2 METHODS

2.1 Task, body and world

The body is spider_8 from ARIEL’s John Set: a core with four legs of two hinges each, eight position-controlled motors in total. The world is the OlympicArena: a flat start, then from $x = 0.5$ m a strip of centimetre-high bumps on which the target lies at (2, 0) m (Figure 1). The robot spawns at the origin with ARIEL’s default spawn, and each walk lasts 15 s, as in the template; it ends early when the core comes within 0.1 m of the target. The network is queried at 50 Hz (every 10 physics steps of 2 ms); the motors’ servos move the joints in between.

Both body and world are free choices that stay fixed for the whole study; we chose them in pilots. Our first body, spider_16, cannot lift itself: holding its 4 kg up needs about 5 N m per leg, while every John Set motor is limited to 0.66 N m. On the rugged world the mean slope along the path is 25° and no body we tried got far; on the flat world spider_8 solves the task within about 2,000 evaluations, leaving no room for conditions to differ. The OlympicArena is learnable but not trivial.

ARIEL draws the bumpy strip with unseeded noise, so every build is a new arena. Each seed therefore builds one arena once and every condition with that seed walks the same saved arena. This makes the fitness deterministic and the comparison between conditions paired by seed. Generalisation is tested afterwards on 20 fresh arenas (Section 3).

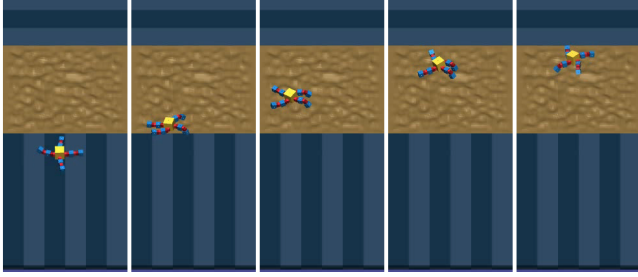


Figure 1: An evolved brain (standard EA, seed 4) walking up the OlympicArena, seen from above, at 0, 4, 8, 12 and 14.3 s. It starts on the flat part (bottom) and reaches the target on the bumpy strip.

Table 1: The controller’s 16 inputs, all scaled to about $[-1, 1]$.

Input	Size	What it tells the controller
Joint angles	8	where each hinge is now
Clock	2	sin and cos of a 1 Hz beat, for rhythm
Target	3	distance / 2 m; sin, cos of its bearing
Tilt	3	the up direction seen from the core

2.2 Controller

The controller is a fully connected feed-forward network with two hidden layers of 8 and 4 tanh neurons and 8 tanh outputs, each scaled to a target angle in $[-\pi/2, \pi/2]$ for one hinge. With a bias on every neuron it has 212 weights, which form the genome. Its 16 inputs are listed in Table 1. The network has no memory, so it gets a clock to walk to; the clock is a fixed input signal, not an oscillator inside the controller, so the controller is not a CPG. The target is given relative to the robot’s heading rather than as an absolute position, so that “the goal is 1 m away, 30° to the left” means the same everywhere on the arena.

Why no vision. Our first controller also had ten distance rays to the terrain: one down, one up and two per face of the core, so that each side could “see” the ground ahead. They were designed for the rugged world, whose hills are up to 0.5 m high. On the OlympicArena only five of them carried terrain information (the up ray never changed and the four shallow rays mostly measured the arena’s edges), so we kept those five, and then tested three, one and none against them (three tuning seeds). No smaller set beat five rays on every seed, and no vision tied with them (0.97 against 0.92 m from the target on unseen arenas; training fitness 1.339 against 1.348). Measuring the signals explained why: a ray reads up to 3 m while the bumps are about ± 2 cm, so a bump moves a ray input by less than 0.01, 20–40 times less than the joint angles move, and the rays vary almost as much on flat ground: they mostly sensed the robot’s own bobbing. We therefore removed vision, which also shrank the genome.

Why 16-8-4-8. Six shapes from 208 to 808 weights were compared on three seeds (Table 2). The bottleneck of four was the only shape better than our previous single layer of 16 on every seed (mean best fitness 1.203 against 1.339 at 6,000 evaluations) and also

the best on unseen arenas (0.85 against 0.97 m). Driving eight motors through four shared signals pushes the legs towards coordinated patterns, and halves the number of weights to search; wider networks were clearly worse at this budget.

2.3 Fitness function

The assignment’s fitness is the planar distance d_T between the core’s final position and the target. We started with exactly that, and nothing walked: the best networks of our first runs wiggled in place, their joint commands within ± 0.13 of their range. Penalising the core for touching the ground and for being upside down made the robots move, but on video they crouched (the core 1.6 cm above lying height, although the body can hold it 6 cm up) and kept one leg tucked away ($5\text{--}9^\circ$ of movement against $17\text{--}77^\circ$ for the other legs). The distance alone gives evolution no reason to walk well before it walks at all, so we added terms for the mechanics of walking, and minimise

$$f = d_T + 0.5 \bar{d} + 1.0 c + 1.0 u + 1.0 \ell + 0.5 w, \quad (1)$$

where every term is measured at each 50 Hz update and lies in $[0, 1]$ except the distances:

- **Stand** (c, ℓ): c is the share of the walk in which the core touches the ground, and ℓ is the mean shortfall of the core’s lift above its lying height from a 4 cm line, $\text{clip}(1 - \text{lift}/4 \text{ cm}, 0, 1)$. The body can hold itself 6 cm up, so 4 cm is demanding but reachable, and every millimetre up to the line pays.
- **Use all legs alike** (w): just as people use both legs about equally, each leg should do about a quarter of the work. With W_i the positive motor work of leg i , $w = 1 - \min_i W_i / \bar{W}$: 0 when every leg works alike, 1 when one leg is idle. We first measured how much each leg moves, but a leg can move without pushing, and a leg that only props the body up carries its share of the weight; only the motor work caught such a lazy leg (22 J against up to 40 J for the others).
- **Stay upright** (u): the share of the walk spent upside down.
- **Be fast** ($\bar{d}$): the distance to the target averaged over the whole 15 s, counting 0 after arrival, so that closing the distance early, and arriving early, lower the fitness.

The third mechanic of walking, **rhythm**, is not a fitness term: it is provided by the clock input. A robot that holds still scores 5.0 ($d_T = \bar{d} = 2$, core on the ground, nothing lifted), and one that arrives at once while standing tall and using all legs alike scores 0. A typical evolved brain (no migration, seed 0) scores $0.15 + 0.5 \cdot 1.19 + 0.02 + 0 + 0.18 + 0.5 \cdot 0.19 = 1.04$: the two distance terms dominate, and the gait terms decide between walkers that get about equally far. This is behavioural shaping: it prescribes part of *how* to walk, which the assignment allows if the metric is clearly defined. All conditions use the same fitness, and we report the plain distance as well.

2.4 The evolutionary algorithm

The genome is a fixed-length vector of real numbers, so we use a real-coded genetic algorithm: Gaussian mutation explores around good weights, and recombination can combine good neurons found in different individuals. The island model is both the object of our

study and a known guard against premature convergence on multimodal landscapes.

Representation and variation. A genome is the vector of 212 real weights, initialised from $\mathcal{N}(0, 0.5^2)$. Parents are chosen by tournaments of three within an island [6]. With probability 0.9 a child is made by *neuron-level uniform crossover*: for every hidden neuron, a fair coin decides from which parent it inherits all of that neuron’s incoming weights and bias (the second hidden layer’s neurons also take their outgoing weights); each output bias comes from a random parent. Otherwise the child copies one parent. Every child is then mutated by adding $\mathcal{N}(0, 0.05^2)$ noise to every weight, with ARIEL’s Gaussian mutation operator. Keeping neurons whole is the classic response to the competing-conventions problem [9, 10]: the child inherits working units rather than a mixture of two units that may play different roles.

Survivor selection. Generational with elitism: each island keeps its two best individuals unchanged and replaces the other 18 by children. As the arena is fixed, the fitness is deterministic and elites are not re-evaluated.

Island model. Four islands of 20 individuals form a ring. Every 10 generations each island copies two emigrants to the next island, where they replace that island’s two worst individuals. The emigrant-selection policy is the variable of the RQ: *best* (the two fittest), *worst* (the two least fit), *random* (two chosen uniformly) or *none* (no migration, four isolated islands). Everything else is identical across conditions.

Baselines. The *standard EA* is the same algorithm with one population of 80 and 8 elites, so only the population structure differs. *Random search* draws every new genome from the initial distribution and keeps the best, at the same budget, as the assignment requires.

Budget. Each run stops after 12,000 evaluations (80 initial and 72 children per generation: about 166 generations). Instead of a plateau rule, which never fired before the budget ran out in our pilots, we chose the budget from where the pilot curves flattened: 95% of the progress of a long run came within 5,000 evaluations and its last improvement at 8,800.

2.5 How the settings were chosen

Table 2 lists every setting with its justification. Settings were changed by controlled comparisons with a decision rule fixed beforehand: a new value replaces the current one only if its best fitness after 6,000 evaluations is better on all three tuning seeds and on the mean. The tuning seeds (100–102) and their arenas are disjoint from those of the final experiment (0–4), and tuning used the random emigrant policy, which favours none of the policies we compare. The migration interval, the number of migrants, the topology and the replacement were not tuned: they set the context of the RQ and take common values from the literature [4]. Only one tuned value changed: crossover probability 0.9 (mean best fitness 1.369 against 1.800 at 0.5). In total the project used 1.87 million evaluations: 0.80 million for development, 0.53 million for tuning and 0.54 million for the final experiments.

3 EXPERIMENTAL SETUP

Conditions. Six conditions: the island EA with the *best*, *worst*, *random* and *none* policies, the standard EA and random search. Each runs on seeds 0–4; a seed fixes the EA’s random numbers and the arena, and all conditions with the same seed share that arena, which pairs the runs.

Metrics. Lower is better for all. (1) The best-so-far fitness against evaluations, and its area under the curve (AUC), which rewards being good early. (2) Convergence speed: the evaluations until the best fitness first drops below 1.6. This threshold separated fast from slow runs in tuning, where no untuned run reached it. (3) The final best fitness after 12,000 evaluations, and the closest any brain got to the target. (4) Generalisation: each run’s best brain walks 20 fresh arenas it never saw, and we report the mean distance left.

Statistics. A Friedman test across the six conditions, blocked by seed, asks whether they differ at all; planned two-sided Mann–Whitney tests against *none* with Holm’s correction ask which do. With five seeds the smallest attainable p is 0.008, so only large effects can be significant. We therefore also report, for every pair of conditions, the probability that one has the lower true mean, from a Bayesian paired t -test with a flat prior [2], with differences within ± 0.05 counting as practically equal, and the Vargha–Delaney A_{12} effect size [15].

Follow-ups. (a) The migration interval: the *best* policy migrating every 5, 20 and 50 generations, on the same seeds and arenas; the main experiment supplies every 10 and never. (b) Longer test walks: every best brain walks its own arena for 15, 20, 30 and 60 s and the unseen arenas for 30 s, to see whether the 15 s limit or the gait stops the robots.

Reproducibility. The code builds on ARIEL without changing it; one script per experiment runs all seeds of all conditions. A run takes about 10 minutes on an Apple M3 Pro with 10 worker processes; the 45 final runs took 6.4 hours. **[TODO: name the scripts or the repository in one sentence]**

4 RESULTS

4.1 Convergence and final fitness

Figure 2 and Table 3 summarise the main experiment. Every EA beats random search by a wide margin (Friedman $\chi^2 = 13.0$, $p = 0.023$; random search against no migration, Holm-corrected $p = 0.04$; posterior probability 100% for every EA). Random search never reached a fitness of 1.6 and ended 1.32 m from the target.

The policies differ in *speed*. Migrating the best reached fitness 1.6 after $2,906 \pm 364$ evaluations, the other policies after about 4,900 and the isolated islands after $5,466 \pm 2,329$. Measured by the AUC, every migration policy is faster than no migration with posterior probability 89–96%, and migrating the best is faster than migrating the worst (88%) or random individuals (68%). The policies hardly differ in the *outcome*: after 12,000 evaluations best and random are practically tied (1.130 against 1.137; 52%), and both are probably, but far from certainly, better than no migration (73% and 84%; e.g. best minus none -0.115 , 95% interval -0.580 to $+0.351$). None of the planned frequentist comparisons between EAs is significant. Unexpectedly, the standard EA ended best (0.957 ± 0.252 ; 78–94%

Table 2: Every setting of the final EA and how it was chosen. “Tested” rows were decided by the rule of Section 2.5 (3 seeds, 6,000 evaluations); best fitness shown as the mean over the seeds, lower is better.

Component	Setting	How it was chosen
Network	16-8-4-8, tanh, 212 weights	tested against 16-8-8, 16-16-8, 16-32-8, 16-8-8-8, 16-16-16-8: only shape better on every seed (1.203 vs 1.339)
Inputs	16, no vision	10 → 5 rays (pilot); 5 vs 3, 1, 0 rays tested: no difference, rays carry <0.01 of signal
Parent selection	tournament, $k = 3$	$k = 2$ and 5 tested: inconsistent (1.877, 1.612 vs 1.800)
Crossover	neuron-level uniform	tested against weight-level uniform (1.416), BLX-0.5 (1.168, better on 2 of 3 seeds) and headless chicken (2.207) vs ours (1.203)
Crossover probability	0.9	0, 0.5 and 0.9 tested: 0.9 better on every seed (1.369 vs 1.800; 0 gave 2.063)
Mutation	Gaussian, $\sigma = 0.05$, every weight	pilot over $\sigma \in \{0.05, 0.1, 0.2\}$; then 0.02, 0.1, and 10% of weights with $\sigma = 0.16$ tested: none better
Mutation-step control	none	a rule doubling σ on stagnating islands tested: 1.369 vs 1.348 without, dropped
Survivor selection	generational, 2 elites per island	1 and 5 elites tested: inconsistent (1.791, 1.811 vs 1.800)
Population	4 islands $\times$ 20	4×10 and 4×40 tested: inconsistent (2.020, 1.693)
Migration	ring, every 10 generations, 2 copies, replace worst	literature [4]; not tuned (context of the RQ); interval studied afterwards (Section 4.3)
Budget	12,000 evaluations (≈ 166 generations)	pilot curves: 95% of the progress by 5,000, last gain at 8,800
Walk	15 s, ends at the target	the template’s length; 20 s training walks tested: no better on unseen arenas

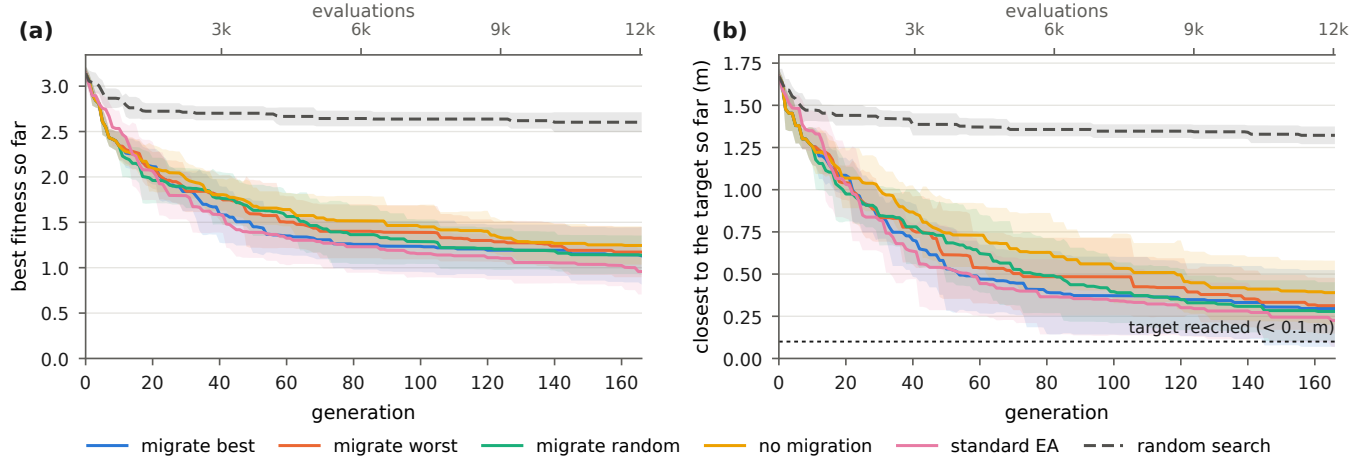


Figure 2: Best-so-far fitness (a) and closest distance to the target (b) against generations (top axis: evaluations), for every condition: the mean over 5 seeds, with a band of ± 1 standard deviation. Lower is better.

Table 3: Results after 12,000 evaluations, mean $\pm$ standard deviation over 5 seeds (lower is better). “Closest” is the closest any brain of the run got to the target; “to 1.6” counts only the runs that got there. P(faster) is the posterior probability that the condition has a lower AUC than no migration.

Condition	Final fitness	Closest (m)	Evaluations to 1.6	AUC	Unseen arenas (m)	P(faster than none)
Migrate best	1.130 $\pm$ 0.317	0.296 $\pm$ 0.227	2,906 $\pm$ 364 (4/5)	1.491 $\pm$ 0.218	0.857 $\pm$ 0.250	96%
Migrate worst	1.172 $\pm$ 0.277	0.313 $\pm$ 0.165	4,976 $\pm$ 2,626 (5/5)	1.582 $\pm$ 0.226	0.830 $\pm$ 0.178	94%
Migrate random	1.137 $\pm$ 0.217	0.279 $\pm$ 0.173	4,846 $\pm$ 1,797 (5/5)	1.545 $\pm$ 0.224	0.863 $\pm$ 0.078	89%
No migration	1.245 $\pm$ 0.211	0.390 $\pm$ 0.190	5,466 $\pm$ 2,329 (5/5)	1.649 $\pm$ 0.164	0.911 $\pm$ 0.156	–
Standard EA	0.957 $\pm$ 0.252	0.225 $\pm$ 0.157	3,190 $\pm$ 1,436 (5/5)	1.431 $\pm$ 0.207	0.885 $\pm$ 0.107	90%
Random search	2.603 $\pm$ 0.110	1.322 $\pm$ 0.052	never (0/5)	2.681 $\pm$ 0.072	1.454 $\pm$ 0.108	0%

better than each island variant) and was about as fast as migrating the best (AUC 65% in its favour). Figure 3 shows every pairwise comparison of convergence speed.

4.2 What the fitness rewards

Figure 4 splits the champions’ fitness into its terms. The speed term $0.5 \bar{d}$ is the largest part of every EA’s champion, about half of its

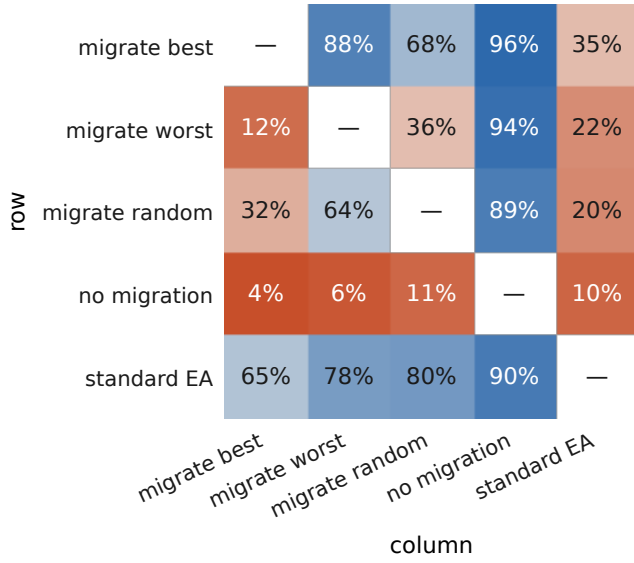


Figure 3: Probability that the row condition converges faster (has a lower AUC) than the column condition, from a Bayesian paired t -test over the 5 seeds. Blue: the row is probably faster; orange: probably slower.

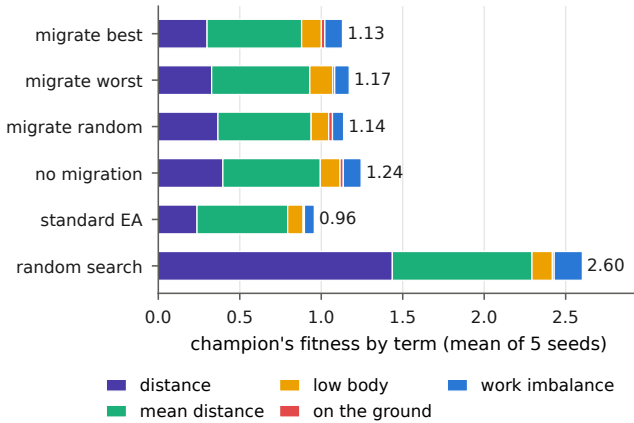


Figure 4: The final champions' fitness split into the weighted terms of Equation 1, mean over 5 seeds; "mean distance" is 0.5 $\bar{d}$. No champion was ever upside down.

fitness, and larger than the final distance. The gait terms add only 0.20–0.27 together: no champion is ever upside down and the core hardly touches the ground ($c \leq 0.02$), so the remaining gait differences come from body height and work balance. The posture terms shaped the gait early in evolution and then left the ranking to the distance terms.

4.3 Generalisation to unseen arenas

On their own arena the best brains end 0.23–0.39 m from the target; on 20 unseen arenas they end 0.83–0.91 m away, whatever the policy (Table 3). The differences between the EAs there are

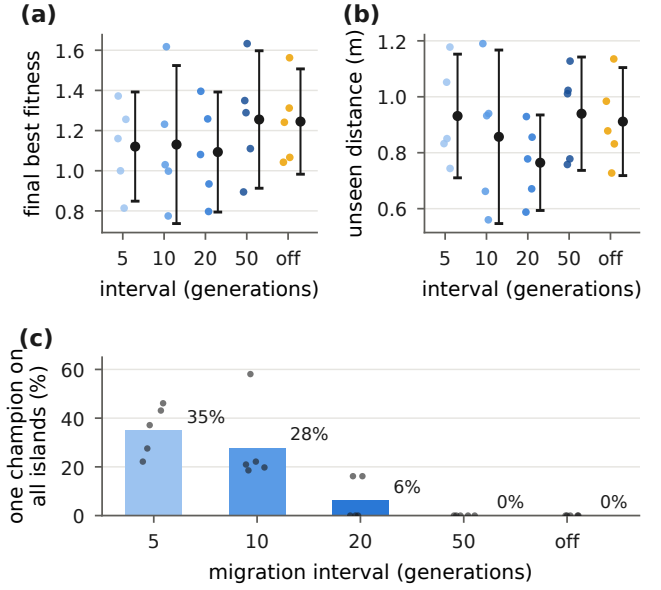


Figure 5: The *best* policy migrating every 5, 10, 20 or 50 generations, or never ("off"), 5 seeds each. (a) Final best fitness and (b) distance left on 20 unseen arenas: dots are seeds, black the mean with its 95% interval. (c) Share of generations in which all four islands hold the same champion; dots are seeds.

Table 4: Best brains that reach their own target when walking longer, of 30 (5 seeds $\times$ 6 conditions), and the share of unseen-arena walks that reach it.

Walk length	15 s (training)	20 s	30 s	60 s
Own arena	3	11	11	12
Unseen arenas	0–4%	–	5–13%	–

practically zero: for best against worst, 61% of the posterior lies within ± 0.05 m. All evolved brains specialise to the arena they were trained on.

4.4 Follow-up: how often to migrate

Figure 5 shows a clear sweet spot. Migrating every 5, 10 or 20 generations is equally fast (AUC probabilities 42–58% between them), and all three are faster than every 50 generations (93–99%) or never (96–100%). Three migrations in 166 generations are too few to matter: every 50 is as slow as no migration. The more often the best migrate, the sooner one champion occupies all islands: in 35% of generations at every 5, against 6% at every 20. Every 20 generations kept the islands apart and still had the best final fitness (1.093) and the best result on unseen arenas (0.76 m; 77% better than every 10, 99% than never).

4.5 Follow-up: is 15 seconds enough?

Only 3 of the 30 best brains reach the target within the 15 s they were trained for (Table 4, Figure 6). Given 20 s, eight more arrive,

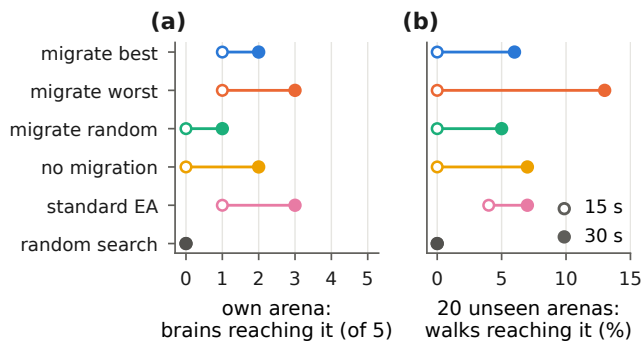


Figure 6: Trained on 15 s walks, tested for 15 s (open) and 30 s (filled). (a) Best brains that reach the target on their own arena, of 5 per condition. (b) Share of the walks on 20 unseen arenas that reach it.

between 15.2 and 19.6 s: they head for the target and run out of time. The others stall 0.2–0.7 m away, legs still moving but the body no longer advancing, and more time does not help (one more arrives within 60 s). On unseen arenas, 30 s instead of 15 s raises the share of walks that arrive from 0–4% to 5–13%; generalisation remains the main limit. Random search never arrives, and the policy hardly matters here either (1–3 of 5 brains per EA at 30 s).

5 DISCUSSION

Answer to the RQ. The emigrant-selection policy changes how fast the islands improve but not where they end. H1 is supported: any migration beats isolated islands on convergence speed (89–96%). H2 is half supported: migrating the best is the fastest, as the selection-pressure argument predicts [4], but within our budget it does not end at a worse plateau than random migration; the two tie. H3 is rejected: migrating the worst still speeds up the islands nearly as much as random migration (94% faster than none).

Why migrating the worst helps. In a generational EA with two elites per island, an island’s worst individuals are not random genomes: they are fresh children of tournament-selected parents, carrying the source island’s working neurons. With a crossover probability of 0.9, the receiving island recombines these neurons with its own, so even the worst emigrants inject useful diversity. Moreover, every policy replaces the receiver’s worst individuals, so even the worst emigrants only displace natives that are no better.

Diversity against speed. Migrating the best quickly copies one champion around the ring: at every 10 generations all four islands share it in 28% of the generations, from about 6,800 evaluations on. That explains its speed, and the follow-up shows the cost: migrating less often (every 20) kept the islands different and gave the best final and unseen results, while migrating too rarely (every 50) wasted the islands entirely. Within this design, the interval is the stronger lever; the policy decides mainly how fast good genes spread.

Why one population did as well. Islands trade speed for diversity that pays off in long runs on deceptive problems. Our budget is short (166 generations) and the panmictic population keeps the 8

best individuals overall, where the islands keep the 2 best of each island, some of which are worse. Exploitation wins at this budget; a longer run might reverse the order. [TODO: decide whether to test this or leave it as future work]

Limitations. Five seeds give wide intervals: most differences between policies are probable rather than certain, and none is significant by a classical test. The settings were tuned with one factor at a time on three seeds, which misses interactions [5], and with the random policy, so they may suit the policies unevenly. The fitness shapes the gait (Section 2.3), so “better” means better by our definition; the plain distance gives nearly the same ordering (Table 3). Brains specialise to their arena, a single body and world were used, and the 15 s limit stops a third of the brains that would arrive. Training on three arenas with turned starts, or with 20 s walks, did not improve generalisation in a pilot at 6,000 evaluations [8]; it may need a larger budget.

6 CONCLUSION

We evolved neural controllers that make `spider_8` walk to a target 2 m away and asked whom an island model should send between its islands. Migration of any kind speeds up evolution compared with isolated islands, and migrating the best is the fastest, reaching a fitness of 1.6 in about half the evaluations of isolated islands. The final controllers and their behaviour on unseen arenas do not depend measurably on the policy, and a single population did as well at our budget. How often the islands migrate matters more: every 5–20 generations is equally fast, every 50 is as slow as never, and every 20 kept the islands most diverse. Future work: longer budgets, where the diversity islands preserve should pay off; more seeds; recombination that aligns an immigrant’s neurons with the natives’ before crossing [14]; and training that varies the arena enough for the controllers to generalise. These conclusions rest on five seeds, one body, one world and a fitness that shapes the gait.

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CONTRIBUTIONS

[TODO: each member: one or two lines on what they did]